

Research paper

Comparison of various machine learning techniques for modeling the heterogeneous acid-catalyzed alcoholysis process of biodiesel production from green seed canola oil

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ABSTRACT

Multiple machine learning (ML) algorithms were developed using artificial intelligence, including Linear Regression (LR), Decision Tree (DT), Random Forest (RF), and K-Nearest Neighbor (KNN), to predict the yield of biodiesel production in an acid-catalyzed alcoholysis process using green seed canola oil. Catalyst loading, methanol-to-oil (M/O) molar ratio, and reaction time were considered as input parameters, while the yield of biodiesel production was selected as the output parameter. The performance of the developed ML models was assessed using evaluation metrics such as the coefficient of determination (R^2) and the root mean squared error (RMSE). The R^2 values obtained for LR, RF, DT, and KNN models were 0.80, 0.95, 0.97, and 0.84, respectively. Furthermore, the corresponding RMSE values for these models were 2.48, 1.51, 0.89, and 4.51, respectively. According to the results, the DT model exhibited superior accuracy and reliability for predicting biodiesel production compared to the other models. The values of the input variables to potentially yield the highest biodiesel output were identified through a systematic trial-and-error approach using the DT model. The results showed that a biodiesel yield of 88 % can be achieved with 5 wt% catalyst loading, a 22 M/O molar ratio, and a reaction time of 5 hours.

1. Introduction

Fossil fuels are the main supplier of the world's energy requirements, demonstrating society's heavy dependence on finite nonrenewable resources (Aghbashlo et al., 2021). Global energy consumption is predicted to increase by 56 % during 2010–2040. Along with the limited resources of fossil fuels, their consumption as the main resource of energy has led to many environmental and health concerns. This, therefore, shows an urgent need for a sustainable alternative energy resource. There has been an increase in the popularity of biofuels worldwide as they benefit the environment by considering their clean, sustainable, and biodegradable properties (Esmi et al., 2021).

Biodiesel, a sustainable energy source, shares key characteristics with conventional petroleum diesel, including high combustion efficiency, a high flash point, and an elevated cetane number. Its environmentally friendly attributes make biodiesel a viable green energy alternative, sourced from either plant oils or animal fats. Various

approaches have been utilized in the production of biodiesel, including pyrolysis, micro-emulsion, and alcoholysis. Among these methods, alcoholysis, which is also known as esterification or transesterification reaction, has emerged as the predominant method due to its ability to yield biodiesel of superior quality (Gebremariam and Marchetti, 2017).

In order for biodiesel to be economically feasible, the choice of an appropriate raw material is critical in terms of being locally available, low-cost and non-edible (Demirbas et al., 2016). Green seed canola oil is an economically beneficial oilseed that is abundantly available in Canada and is not suitable for edible purposes. Homogenous (acid/base) or heterogeneous (acid/base) catalysts can be used in transesterification reactions, however, heterogeneous acid catalysts are preferable as they tolerate free fatty acids present in low-quality oils and due to their benefits including easier separation, and higher reusability (Esmi et al., 2022a).

Biodiesel production through the alcoholysis process has attracted significant interest and numerous scientists have worked in this area

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(Zhang et al., 2023; Kingkam et al., 2022; Ma et al., 2021). The majority of studies in this field aim to improve the yield of biodiesel production by evaluating the process experimentally, developing novel catalysts, and optimizing the process (Basumatary et al., 2022). In terms of yield and process costs in biodiesel production, the transesterification process plays a significant role. It is therefore crucial to optimize the reaction conditions to maximize biodiesel yield (Abdelbasset et al., 2022a).

It has become increasingly popular to use machine learning methods for modeling the biodiesel production process and reducing production costs. Optimization and evaluation of biodiesel yield using machine learning (ML) models requires the least experimental tests and is cost-effective. ML is a branch of artificial intelligence in which models are trained by imported experimental data to manage complex tasks (Nguyen et al., 2021). ML has been employed widely by many researchers in recent years to predict and optimize the efficiency of the biodiesel process (Kumar et al., 2022; Almohana et al., 2022; Ame-naghawon et al., 2021; Chen et al., 2022). The statistical techniques of machine learning can analyze the impact of individual input factors as well as the interaction between the input factors on yield. The choice of an appropriate modeling approach is essential for optimizing the process toward achieving maximum biodiesel yield (Sun et al., 2022). The integration of control algorithms and machine learning techniques in biodiesel production can enhance process efficiency, improve product quality, and contribute to overall sustainability and cost-effectiveness (Mondal et al., 2022). Control algorithms can help optimize the biodiesel production process by continuously adjusting parameters to achieve the desired product characteristics and maximize yield (Ahmad et al., 2022). Various ML methods have been employed in the biodiesel process such as Linear Regression (LR) (Abdelbasset et al., 2022b), Artificial Neural Network (ANN) (Ayoola et al., 2019), K-nearest Neighbor (KNN) regression (Jin et al., 2022), Random Forest (RF) (Abdelbasset et al., 2022b), Decision Trees (DT) (Sun et al., 2022), and Support Vector Regression (SVR) (Filgueiras et al., 2014).

Maulud et al (Maulud and Abdulazeez, 2020). reported a comprehensive review of the LR model in machine learning, exploring the contributions of various researchers in the field of linear Regression and polynomial regression. This study focuses on evaluating the precision and prediction capabilities of these models in order to optimize their performance. Abdelbasset et al (Abdelbasset et al., 2022b). employed multiple algorithms, including the Boosted Linear Regression model, to improve the production of heterogeneous catalytic biodiesel from waste vegetable oil. The R^2 score and error rate of 0.93×10^{-2} were reported for this model.

ANN is a powerful and effective method for modeling nonlinear multifactor systems. Biodiesel production efficiency can be predicted with the help of an ANN, which has the capability of learning data that have complicated behavior (Sarve et al., 2015). SVR is a supervised regression model, which has been commonly employed for predicting non-linear complex processes (Fernández-Delgado et al., 2019). Three mathematical methods including SVR, ANN, and Response surface methodology (RSM) have been employed to predict the yield of biodiesel obtained from papaya seed waste oil. The accuracy of models was assessed, which showed that SVR model fits the experimental data better than others with lower relative error and higher correlation coefficient (Sultana et al., 2019). DT models are based on a tree-like structure, in which the data are arranged under a root node containing several branches and the leaf nodes represent the output. Cosgun et al (Cosgun et al., 2021). investigated the productivity of *Yarrowia lipolytica* for biodiesel synthesis using DT approach and found DT model as an effective approach for predicting optimum conditions for biodiesel production.

KNN algorithm as one of the most straightforward models in ML, predicts the relationship between the unseen data and existing data to assign the new data to the most related category (Sun et al., 2022). Gautam et al (Gautam et al., 2022). developed different machine learning models, including LR, KNN, and MLP, to predict the optimal

synthesis of biodiesel from pre-treated and non-treated Linseed oil through base transesterification reaction. The R^2 values obtained for LR, KNN, and MLP were 0.70, 0.85, and 0.79, respectively. These results indicate that the KNN model surpasses the other models in terms of accuracy, displaying a higher level of precision and a lower error rate. Table 1 presents a summary of the literature review on the applications of different machine learning algorithms in the biodiesel production process.

The present study investigates various statistical approaches of LR, DT, RF, and KNN for simulation and optimization of biodiesel production obtained from unrefined green seed canola oil through the heterogeneous acid-catalyzed-alcoholysis process. There is no study comparing LR, DT, RF, and KNN algorithms for modeling biodiesel production. The data used in this study to model the yield of biodiesel from unrefined green seed canola oil using 15 % 12-tugstophosphoric heteropoly acid immobilized on mesoporous aluminosilicate catalyst (HPW/MAS) has not been studied before using machine learning approaches. The primary aim was to assess the impact of changes in reaction conditions (catalyst weight, methanol to oil molar ratio, and reaction time) within an acceptable range as input variables, in order to predict the highest biodiesel yield as the output variable. The accuracy of the predicted models was evaluated with different performance analyses and compared to find the best-fitted model.

2. Data set

For this study, 27 experimental data points were collected (Esmi et al., 2022a, 2022b) as shown in Table 2. Data were obtained by measuring three input variables (catalyst loading, M/O molar ratio, and reaction time) within a suitable range (Hidayat and Sutrisno, 2018; Pan et al., 2018; Zanette et al., 2011; Pires et al., 2014), and a single output, the yield (%) of biodiesel.

3. Modeling of process

The selection of an appropriate modeling approach is crucial in achieving meaningful and accurate results in machine learning. The first step in this process involves determining the type of ML learning that aligns with the specific task and the nature of available data, whether it's reinforcement learning, supervised learning, semi-supervised learning, or unsupervised learning methods. In supervised learning method, the machine learning algorithm is provided with input parameters presented as data, and the objective is to input known parameters into the algorithm and predict unknown output parameters. Given the nature of

Table 1
Comparative analysis of machine learning models in biodiesel production.

Model	Application	Statistical analysis	Reference
Boosted LR	Biodiesel production from waste cooking oil	$R^2=0.93$, error rate=0.057	(Abdelbasset et al., 2022b)
Boosted MLP	Biodiesel production from waste cooking oil	$R^2=0.998$, error rate=0.052	(Abdelbasset et al., 2022b)
RF	Biodiesel production from waste cooking oil	$R^2=0.992$, error rate=0.098	(Abdelbasset et al., 2022b)
KNN	Biodiesel production from waste soybean oil	$R^2=0.99$, RMSE=0.80	(Sun et al., 2022)
DT	Biodiesel production from waste soybean oil	$R^2=0.92$, RMSE=1.86	(Sun et al., 2022)
MLP	Biodiesel production from waste soybean oil	$R^2=0.99$, RMSE=0.49	(Sun et al., 2022)
ANN	Biodiesel production from waste groundnut oil	$R^2=0.93$, R=0.96	(Ayoola et al., 2019)
KNN	Biodiesel production through transesterification process.	$R^2=1$, RMSE=1.09	(Jin et al., 2022)
SVR	Quantification of animal fat biodiesel in soybean biodiesel	$R^2=1$, RMSEP=0.45	(Filgueiras et al., 2014)

Table 2

All the experimental dataset used for modeling the biodiesel production (Esmi et al., 2022a, 2022b).

RUN	X1= Catalyst loading (wt%)	X2= Methanol: oil molar ratio	X3= Reaction time (h)	Y= Yield (%)
1	3.25	11.25	8	72.9
2	3.25	11.25	4	69.7
3	5.5	17.5	6	89.8
4	5.5	5	6	66.1
5	5.5	17.5	10	83.1
6	10	17.5	6	77.1
7	5.5	17.5	6	81.5
8	7.75	23.75	4	75.6
9	5.5	17.5	6	84.2
10	3.25	23.75	4	82.3
11	3.25	23.75	8	84.1
12	7.75	23.75	8	87.4
13	7.75	11.25	8	74.8
14	7.75	11.25	4	70.8
15	5.5	30	6	87.9
16	5.5	17.5	6	83.8
17	5.5	17.5	2	73.6
18	5.5	17.5	6	79.8
19	5.5	17.5	6	81.0
20	1	17.5	6	71.3
21	3	20	1	69.5
22	3	20	2	70.7
23	3	20	3	74.1
24	3	20	4	76.6
25	3	20	5	78.2
26	3	20	6	79.1
27	3	20	7	82.3

the data in this study, which involves predicting the yield of biodiesel production with a given set of independent variables, the logical choice is to employ supervised learning machine models. In supervised learning, a variety of models are available, each with its own strengths and characteristics (Aghbashlo et al., 2021). Through a comprehensive literature review, several effective supervised learning methods were identified for comparison in modeling biodiesel production, including Decision Trees (DT), Random Forests (RF), Linear Regression (LR), and k-Nearest Neighbors (KNN) algorithms. DT, RF, and KNN were selected for their proven effectiveness in handling nonlinear relationships and complex interactions among variables, characteristic of the biodiesel production process. LR was included to facilitate a comparison between linear and nonlinear relationships among input and output variables in this process. This consideration of different algorithms ensures a thorough exploration of modeling techniques, allowing for the identification of the most suitable model for predicting biodiesel production yield in the process.

3.1. Linear Regression (LR) model

LR algorithm is a fundamental supervised learning model, which is simple and easy to understand. LR is used to predict the dependent parameter based on the independent parameters. LR equation can be expressed according to the below equation:

$$y = \beta_0 + \beta_1 x \quad (1)$$

Where x shows the independent parameter and y shows the dependent parameter of the model. β_0 and β_1 show the y-intercept and slope of the regression line, respectively (Rong and Bao-Wen, 2018).

Linear regression is a commonly used modeling, which is often chosen for modeling biodiesel production when the relationships between the input variables and biodiesel yield can be reasonably approximated as linear. This method is favored for its simplicity and interpretability, making it a suitable choice when the underlying relationships are not highly complex.

3.2. Decision Trees (DT) model

DT technique is categorized as a supervised machine learning algorithm, which can be used in classification and regression problems. This technique is based on the binary tree, which is an important data structure in computer science. DT technique breaks down a large, complicated problem into small simple ones resulting in easier comprehension of the solution. This technique has a clear tree-like structure consisting of non-leaf nodes (decision nodes) and leaf nodes (Fig. 1). Decision nodes contain various branches and represent conditions, while leaf nodes represent predictions, which is the output. Using the decision tree algorithm, a model can be trained to predict fresh datasets via a continuous splitting process (Bansal et al., 2022).

3.3. Random Forests (RF) model

RF is an integrated learning technique that relies on the combination of bagging techniques and random subspace methods and can be used for classification and regression. In this tree-based technique, many decision trees are used, and the target output is determined by averaging the predicted values from the decision trees or by conducting a voting process to employ the most voted value (Fig. 2) RF is an easy method to be implemented with fast training speed, low bias and high variance (Gupta et al., 2021), (Dabiri et al., 2022). RF regression equation is presented as below:

$$y = \frac{1}{K} \sum_{i=0}^K h_i(x) \quad (2)$$

Where K shows the number of trees, x indicates the input vector, and $h_i(x)$ indicates the prediction of the regression trees. The random forest prediction can then be calculated by the mean of the predicted regression trees.

3.4. K-Nearest Neighbor (KNN) model

KNN is a non-parametric algorithm categorized under the supervised machine learning approach. In cases where there is quite a bit of ambiguity in the data set, KNN can be considered the most effective algorithm for segregating it. KNN is designed to classify data points from a database divided into a number of classes based on their proximate neighbors' classes. Classification is the most common application of KNN, but regression problems can also be solved with KNN (Magesh et al., 2022). It is an easy and useful algorithm that can be implemented for assigning new values if any missing values are detected and resampling the data. Based on the prediction of the connection between the unseen data and the existing data, the algorithm assigns the new data to the category that best fits it. In fact, the new data point is sorted according to its neighbors' arrangement. This distance-based algorithm employs the Euclidean distance equation (Eq. 3) for determining the k closest neighbors and selecting the primary class among the k samples.

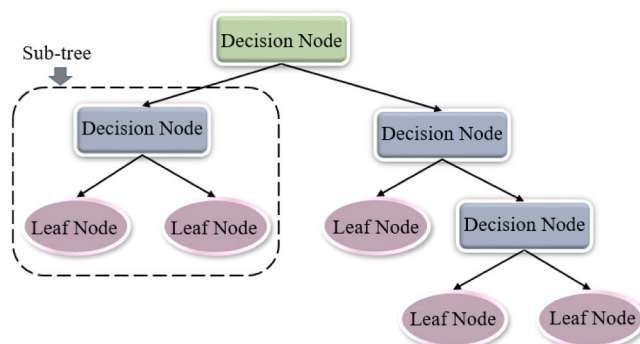


Fig. 1. Decision Trees structure in Machine Learning.

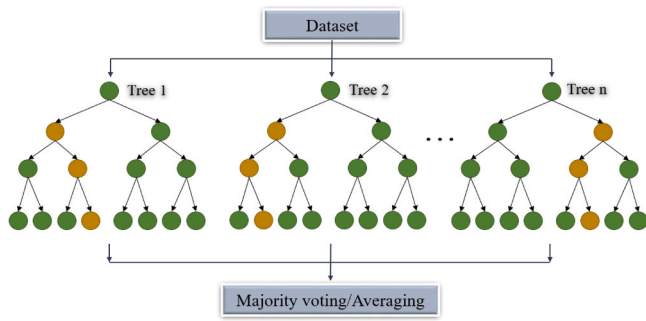


Fig. 2. Random Forest structure in Machine Learning.

In the Euclidean distance equation, x and y are two data points with k dimension. The final output (y) can be obtained by calculating the average of the outputs of its k closest neighbors (Eq. 4) (Gautam et al., 2022).

$$EUC_{Distance} = \sqrt{\sum_{i=1}^k (x_i - y_i)^2} \quad (3)$$

$$y = \frac{1}{k} \sum_{i=0}^k y_i(X) \quad (4)$$

3.5. Accuracy evaluation metrics

Root mean squared error (RMSE) and coefficient of determination (R^2) are important evaluation metrics to investigate the reliability of the models. These metrics are derived from the difference between the predicted and actual values. The smaller the value of these metrics, the higher the accuracy and reliability of the models (Lu and Ma, 2020). RMSE and R^2 can be obtained according to the following equations.

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (x_i - y_i)^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^m (x_i - y_i)^2}{\sum_{i=1}^m (y - y_i)^2} \quad (6)$$

y_i : The observed value

- x_i : The predicted value
- y : Mean value
- m : The total number of observations

Table 3

Evaluation metrics for final models and comparison with other studies.

Model	R^2	R	RMSE	MSE	MAE	MRPD	Max error	Med error	Ref.
LR	0.80	0.89	2.48	6.15	1.94	2.45	8.10	2.07	This work
RF	0.95	0.97	1.51	2.28	1.36	1.72	2.46	1.53	
DT	0.97	0.98	0.89	0.79	0.43	0.52	2.26	0	
KNN	0.84	0.92	4.51	20.34	4.22	5.58	11.04	2.69	
LR	0.52	-	-	5.8	1.80	-	6.20	1.31	Gupta et al., (2021)
RF	0.89	-	1.15	1.33	0.86	-	2.60	0.57	
AdaBoost	0.91	-	-	1.10	0.92	-	1.7	1.05	
AdaBoost Regression Tree	0.99	-	1.55	-	1.22	-	-	-	
Extra Trees	1	-	1.52	-	1.19	-	-	-	Lu and Zhang (2023)
Gradient boosting regression tree	1	-	0.40	-	0.33	-	-	-	
ANN	0.95	-	2.53	-	-	-	-	-	
RF	0.92	-	3.09	-	-	-	-	-	
KNN	0.83	-	6.33	-	-	-	-	-	Miraboutalebi et al., (2016)
SVM	0.87	-	12.12	-	-	-	-	-	
RF	0.95	-	5.18	-	-	-	-	-	
AdaBoost	0.95	-	8.33	-	-	-	-	-	
Boosted MLP	0.99	-	9.81	-	8.38	-	-	-	Sumayli (2023)
Boosted GPR	0.99	-	4.81	-	2.31	-	-	-	
Boosted KNN	0.99	-	6.52	-	5.22	-	-	-	

4. Results and discussion

Evaluation metrics coefficient of determination (R^2), Pearson correlation coefficient (R), root mean squared error (RMSE), mean squared error (MSE), mean absolute error (MAE), mean relative percentage deviation (MRPD), maximum error and median error were used to assess how well the selected models performed and the obtained results are shown in Table 3. Additionally, the evaluation metrics obtained in related research studies are also presented in Table 3 for comparison.

Various hyperparameter combinations were examined to determine the appropriate combinations for DT, RF, and KNN models, as follows: DT:

- Maximum tree depth: 12
- Maximum leaf nodes: 30
- Minimum sample leaf: 2
- Splitter: random

RF:

- Number of estimators: 1400
- Maximum depth: 100
- Minimum samples split: 2
- Minimum sample leaf: 1

KNN:

- Algorithm: brute
- Number of neighbors: 5
- Weights: distance

According to Table 3, R^2 values of 0.80, 0.97, 0.95, and 0.84 were acquired for LR, DT, RF and KNN models respectively, which confirm the better fit of DT and RF models to the experimental data than LR and KNN models. Furthermore, RMSE metric error values of 2.48, 0.89, 1.51, and 4.51 for these models, respectively, indicate the higher reliability of DT and RF models between the developed models.

A study conducted by Liu and Zhang (Lu and Zhang, 2023), evaluated the utilization of palm oil for biodiesel production by employing three ensemble decision tree algorithms: AdaBoost Regression Tree, Extra Trees, and Gradient Boosting Regression Tree. By comparing the RMSE measurements, it was observed that AdaBoost Regression Tree and Extra Trees exhibited an error value of approximately 1.5, whereas the Gradient Boosting Regression Tree model had an error value of around 0.4. Although all the generated models demonstrated robustness, the Gradient Boosting Regression Tree model emerged as the most accurate and reliable in predicting biodiesel output.

Miraboutalebi et al (Miraboutalebi et al., 2016). implemented ANN and RF algorithms to predict the biodiesel cetane number. The ANN

model demonstrated a high level of accuracy with an R^2 value of 0.95 and an RMSE of 2.53. Similarly, the RF model achieved a high level of accuracy with an R^2 value of 0.92 and an RMSE of 3.09. In another study (Gupta et al., 2021), three robust machine learning algorithms: LR, RF, and AdaBoost regression were applied to forecast biodiesel yield. The findings demonstrate that both RF and AdaBoost regression exhibit high accuracy when employed for predictive modeling of biodiesel yield with R^2 of 0.89 and 0.91, respectively. In another study conducted by (Jin et al., 2023), four ML algorithms of KNN, SVM, RF regression, and AdaBoost regression were employed to forecast biodiesel yield from transesterification. Among these models, Random Forest regression stood out as the most accurate predictor, exhibiting lower RMSE values (2.778 for Training and 5.178 for Validation) and a higher correlation coefficient compared to the other models.

In another study conducted by Sumayli (Sumayli, 2023), different models including Gaussian Process Regression (GPR), MLP, and KNN regression, along with adaptive boosting, were used to optimize Papaya oil methyl ester production. Input parameters such as reaction temperature, catalyst quantity, process time, and methanol to oil ratio were considered, while methyl ester yield was the output. Results showed high predictive ability with R^2 scores of 0.988, 0.993, and 0.994 for Boosted MLP, Boosted GPR, and Boosted KNN, respectively. The corresponding RMSE errors were 9.8071, 4.8150, and 6.5180. Optimized methyl ester production of 99.89 % was achieved at specific conditions: 62.5°C temperature, 6.47 minutes reaction time, 0.8125 wt% catalyst, and 10.33 methanol to oil molar ratio.

Fig. 3 illustrates the actual values obtained experimentally versus predicted values obtained using LR (Fig. 3.a), DT (Fig. 3.b), RF (Fig. 3.c), and KNN (Fig. 3.d) approaches for both test and train data. As shown in the test phase of the selected models, the dispersion of data points in DT and RF models are less than in LR and KNN models, which indicates that DT and RF models fit the real data more than LR and KNN models.

Regarding the train phase of the models, DT model demonstrated a better fit to the actual values as compared with LR, RF, and KNN models due to the higher closeness of actual and predicted values. This also was confirmed by the evaluation metrics of R^2 (0.97) and RMSE (0.89) for DT shown in Table 3, which verifies the higher accuracy of DT model than other models. However, based on the R^2 of RF model (0.95), which is slightly less than that in DT model presented in Table 3, and a good fit of actual and predicted values of train data for RF model presented in Fig. 3.c, this model also indicates high accuracy. The training phase in LR and KNN models shows that the data points are more dispersed indicating less reliability of these models compared to DT and RF models. Therefore, according to the results obtained from the test and train phases of data, as well as the results of evaluation metrics, DT model is the most accurate model followed by RF. The accuracy of KNN and LR models was very close (only 0.04 difference in R^2), which can be also observed by comparing their train and test phase diagrams.

Sensitivity analysis was conducted for the Decision Tree (DT) model, identified as the best-fitted model. This analysis involves assessing the impact of changes in input variables on the model's predictions. In this regard, the relative importance of each feature in making predictions was evaluated. Features with higher importance have more influence on predictions. According to the results of the sensitivity analysis, the M/O molar ratio demonstrated the highest importance in predictions, accounting for 54 %, followed by time and catalyst loading with 38 % and 8 %, respectively.

The reaction conditions to potentially achieve the maximum biodiesel yield were attained through exploratory analysis using DT as the best-fitted model. The results showed that the biodiesel yield of 88 % can be obtained with 5 wt% catalyst, methanol-to-oil molar ratio of 22, and reaction time of 5 hours. Almohana et al (Almohana et al., 2022), focused on enhancing the efficiency of biodiesel production from waste cooking oil by utilizing a Fe-exchanged montmorillonite 12 K10 (Fe-MMT K10) heterogeneous catalyst. The research aimed to optimize the production process using three ensemble models of Huber

Regression, Decision Trees, and Gaussian process. These models were boosted using the AdaBoost technique. The R^2 scores achieved for the Boosted Huber Regression, Boosted Decision Tree, and Boosted Gaussian process were 0.814, 0.780, and 0.996, respectively. All the models showed acceptable accuracy, however the Boosted Gaussian process demonstrated superior accuracy compared to the other models in optimizing the production process. By applying specific input values (reaction temperature (°C) =145, reaction time (h) =5.625, catalyst loading (wt%) =4.22, and methanol to oil molar ratio=11.73), the recommended methods yielded an optimal output of 96.75 % biodiesel production.

To assess the accuracy of the obtained results, an experimental analysis was performed using the reaction conditions obtained from the DT model. As a result, a biodiesel yield of 87 % was achieved, demonstrating the high reliability of the model in predicting biodiesel yield, with an error of less than 1.2 %.

Fig. 4 shows the 3D surface plot of the predicted yield of biodiesel obtained by dual effects of input variables. Fig. 4.a indicates that increasing both M/O molar ratio and reaction time led to increasing biodiesel yield. Regarding M/O molar ratio, after the value of 20, a faster trend in the increasing amount of biodiesel yield can be observed. However, there is almost a steady trend in increasing biodiesel yield with an increase in time. In Fig. 4.b, It can be observed that while the increase in both the amount of catalyst and the reaction time led to an increase in biodiesel yield, catalyst loading does not significantly affect the yield. For instance, after 10 hours, increasing the catalyst concentration from 3 to 10 wt% resulted in only about 2 % increase in biodiesel yield. This plot demonstrates that time has a greater impact on biodiesel yield compared to catalyst loading. In Fig. 4.c, the positive impact of increasing the M/O molar ratio and catalyst loading on biodiesel yield can be observed. There is an abrupt increase in biodiesel yield when the M/O molar ratio reaches 20, continuing up to around 22; however, beyond 22, the rate of increase slows down. Additionally, the plot reveals a minor impact of catalyst loading on biodiesel yield with increasing M/O molar ratio, confirming that this parameter has the least impact on biodiesel production.

5. Conclusions

This study investigates the application of four machine learning approaches: Linear Regression (LR), Decision Trees (DT), Random Forest (RF), and K-Nearest Neighbors (KNN), to model the biodiesel production process from unrefined green seed canola oil using HPW/MAS as a solid acid catalyst. The catalyst loading, methanol to oil molar ratio, and reaction time are considered as input factors, while the biodiesel yield is the chosen output factor. All the models demonstrated acceptable accuracy according to the evaluation metrics of coefficient of determination (R^2) and Root Mean Square Error (RMSE), however, the DT model exhibits superior performance in predicting biodiesel yield, with an R^2 value of 0.97 and an RMSE value of 0.89. By using the DT technique, a biodiesel yield of 88 % can be achieved by setting the catalyst loading to 5 wt%, the methanol to oil molar ratio to 22, and the reaction time to 5 hours, respectively. In conclusion, leveraging these cutting-edge machine learning methods is useful for modeling the biodiesel production process, which can potentially lead to reduced production costs.

CRedit authorship contribution statement

Yongfeng Hu: Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Ajay K Dalai:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Fahimeh Esmi:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization.

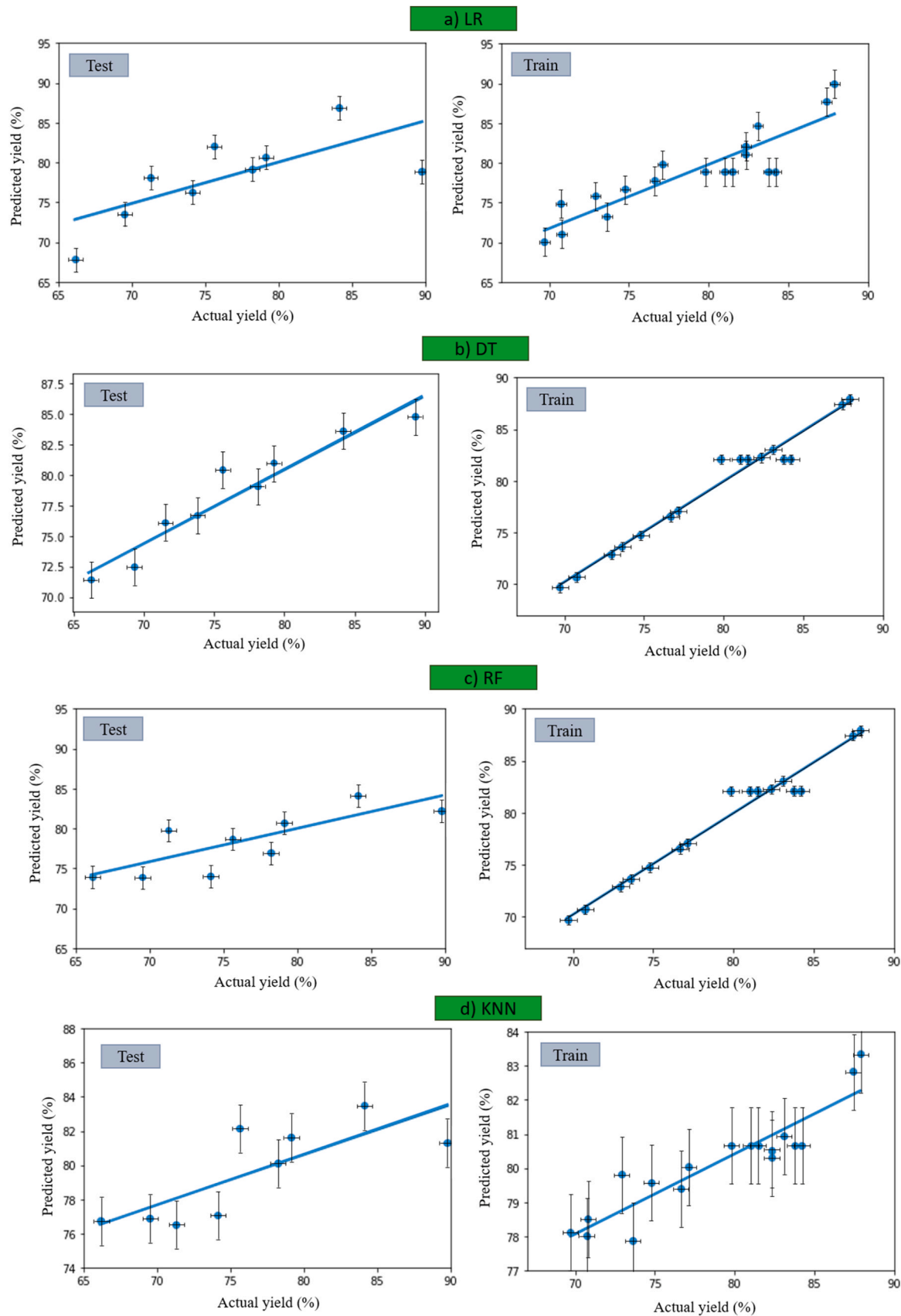


Fig. 3. Test and rain data for correlation between actual and predicted values using a) LR, b)DT, c) RF, and d) KNN approaches.

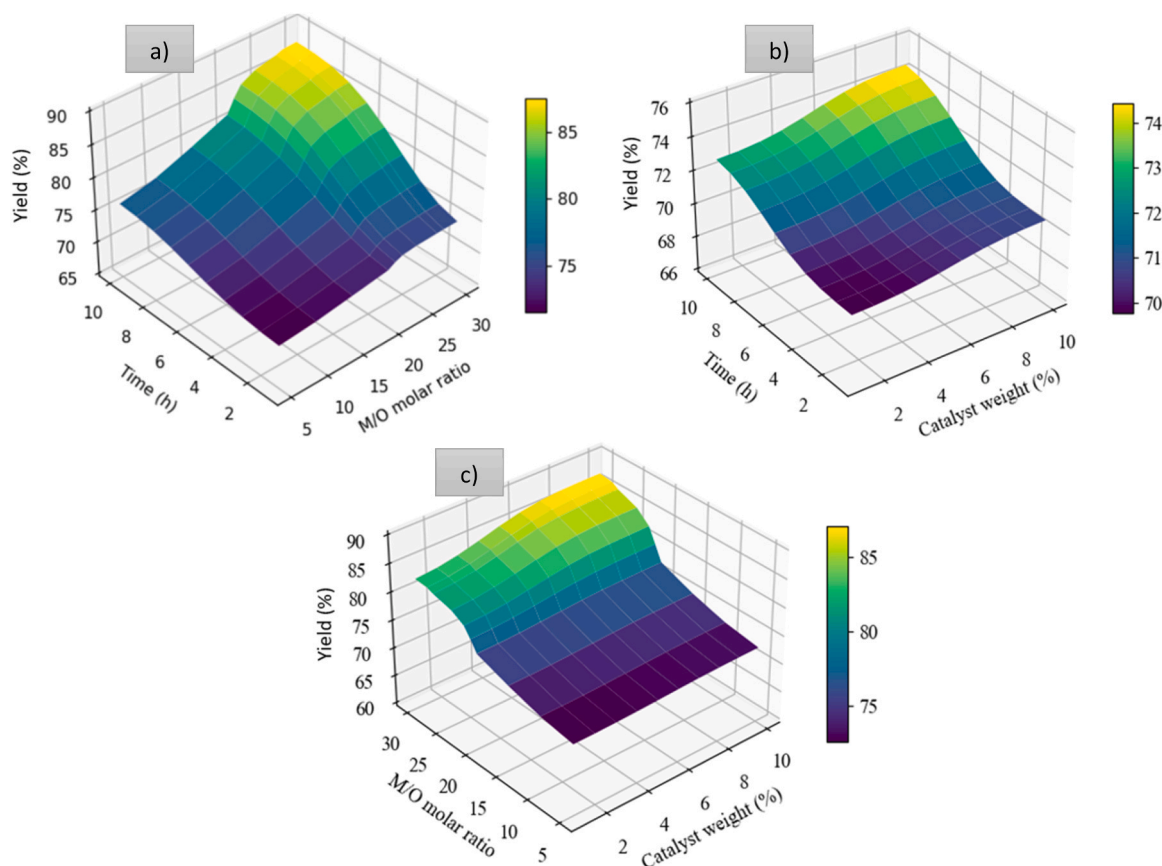


Fig. 4. Dual effect of a) M/O molar ration and time, b) time and catalyst loading (wt%), and c) M/O molar ratio and catalyst loading (wt%) on predicted biodiesel yield by DT model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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